Lee Ngari

SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Software Engineer with internship experience at Microsoft and Aga Khan University Hospital, focused on backend systems, data analytics, cloud engineering, and production automation. Built an analytics pipeline classifying 10,000+ anomalies and cut its runtime by 69%; identified and delivered an on-premises Git and CI/CD platform for a hospital engineering team; built an enterprise healthcare platform and a student product with 2,000+ users. Strong foundation in systems programming and cloud platforms.

EDUCATION

United States International University – Africa

September 2026

B.S. in Software Engineering

TECHNICAL SKILLS

LANGUAGES & FRAMEWORKS	Go, Python, C#, JavaScript, PowerShell, .NET, React, Node.js
DATABASES	PostgreSQL, MS SQL Server
CLOUD, DEVOPS & TESTING	Azure, Docker, Git, Gitea, CI/CD, xUnit, Postman, Grafana, Loki
SYSTEMS & NETWORKING	Linux, Windows, TCP/IP, DNS, HTTP/HTTPS, DHCP, OSI model, CIDR

WORK EXPERIENCE

Microsoft

July 2026 – September 2026

Software Engineering Intern

- Investigated the relationship between live kernel dump (LKD) events and BSOD crashes, identifying independent signals to surface high-impact LKD events from noise.
- Built a post-triage analytics pipeline from scratch that classifies 10,000+ active anomalies using classification, predictive, and statistical models, improving visibility and reducing missed issues.
- Identified and implemented an optimization to the core classification algorithm after deployment, reducing job runtime by 69% and cutting compute and cloud costs.
- Collaborated on Azure resource provisioning and deployment to scale the system in production.

Aga Khan University Hospital

May 2026 – July 2026

Associate Application Developer (Contract)

- Identified the absence of CI/CD and the risk of externally hosted code; pitched and built a self-hosted Gitea server with automated pipelines to on-premises Windows servers, winning manager and team buy-in and full adoption.
- Replaced manual copy-paste deployments that required downtime with automated, faster pipelines that deploy without downtime and support automated rollback.
- Migrated 100+ repositories from GitHub to the on-premises host with zero data loss, removing external code-hosting risk and restructuring branching strategies and access controls to meet internal security policies.
- Validated the new environment with the team and documented the end-to-end migration playbook.

Aga Khan University Hospital

September 2025 – March 2026

Applications Development Intern

- Developed end to end an enterprise claims platform for hospitals and partner insurers, replacing coordination by calls and emails. Supports claim submission, document upload, status tracking, approval or rejection, notifications, and reporting.
- Worked with cross-functional teams to gather and validate requirements; the platform was in early testing with partner insurance companies by the time I left.
- Set up CI/CD pipelines and automated tests, added logging and monitoring (Grafana + Loki), diagnosed production incidents using logs and SQL traces, and documented APIs and workflows.

PROJECTS

Watch Party

Java, Spring Boot

- Built a watch party web application, allowing friends to watch movies together with synchronized playback (play, pause, seek) and live chat over WebSockets.
- Added chunked video streaming with HTTP range requests for fast seeking and smooth playback from local disk storage.
- Integrated WebRTC for peer-to-peer video and audio calling inside watch rooms.

JoyDB: Relational Database Engine

Go

- Built a relational database engine from scratch: lexer, recursive descent parser, AST planner, and executor supporting CRUD, DDL, INNER/LEFT/RIGHT/FULL OUTER JOINs, and foreign key, unique, and auto-increment constraints.
- Designed a Write-Ahead Log with CRC32 checksums, async checkpointing, and crash recovery after ungraceful shutdowns.
- Implemented a B+Tree index that cut insert latency 41% (1,970 to 1,167 ns/op) with ~1 microsecond lookups.

CampusAid — campusaid.co.ke

Azure Functions, Python

- Noticed students relied on friends and group chats for revision material; built and launched a past-paper repository.
- Grew it to 2,000+ users and 5,000+ documents through my own promotion, iterating on the product using user feedback.
- Built Azure Functions for serverless uploads and moderation, with structured logging; automated data cleaning in Python.